

ECON2050 Tutorial 1

Basic Real Analysis I

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Today's Plan

- ▶ Intro
- ▶ Recap: key ideas from Lecture 1
- ▶ Tutorial questions: Q1–Q5, excluding extra practice problems

A bit about Xavier Xi Zhang

- ▶ My name: Xavier Xi Zhang
 - ▶ You can call me **Xavier**, **Xav**, **Xi**, or **X**.
- ▶ Economics Honours student (ECON/MATH student before that).
- ▶ Personal website: xavierxizhang.github.io
 - ▶ Tutorial slides will be uploaded under **Teaching**.
- ▶ Questions are welcome (ECON, MATH, or beyond).
- ▶ My consultation hours are every Thursday from 12:30pm to 2pm, at 39-125A.

- ▶ ECON2050 is regarded as one of the Most Difficult Econ courses!
- ▶ It is a real university course:
 - ▶ The previous courses you took, like ECON1050, are not!
 - ▶ ECON2050 teaches you how to think mathematically and offers a new way of approaching problems.
- ▶ No need to feel anxious, because I'm here to help.
 - ▶ I took this course two years ago and got the highest mark...
- ▶ Want more? I'd be happy to recommend more MATH courses.

Recap: The Space We Work In

Real line

The real line is denoted by $\mathbb{R}$. The distance between $x, y \in \mathbb{R}$ is

$$|x - y|.$$

The distance between $x \in \mathbb{R}$ and 0 is $|x|$.

Euclidean space

More generally, we work in $\mathbb{R}^m$, the set of all m -tuples of real numbers:

$$\mathbf{x} = (x_1, \dots, x_m), \quad m \in \mathbb{N}.$$

When $m = 1$, $\mathbb{R}^m$ is the real line $\mathbb{R}$.

Recap: Norm and Distance in $\mathbb{R}^m$

Norm

For $\mathbf{x} = (x_1, \dots, x_m) \in \mathbb{R}^m$,

$$\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \dots + x_m^2}.$$

The norm is the length of $\mathbf{x}$, or the distance between $\mathbf{x}$ and $\mathbf{0}$.

Distance

For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^m$,

$$\|\mathbf{x} - \mathbf{y}\| = \sqrt{(x_1 - y_1)^2 + \dots + (x_m - y_m)^2}.$$

Recap: Functions

Definition

A function $f: X \rightarrow Y$ assigns to each element of X exactly one element of Y .

Key terms

- ▶ **Domain:** the set X .
- ▶ **Codomain:** the set Y .
- ▶ **Image of x :** the value $f(x) \in Y$.
- ▶ **Range / image of f :** $f(X) = \{y \in Y : y = f(x) \text{ for some } x \in X\}$.
- ▶ **Graph:** $\{(x, y) \in X \times Y : y = f(x)\}$.

Recap: Sequences

Definition

A sequence in X is a function

$$f: \mathbb{N} \rightarrow X.$$

We usually write a sequence as

$$(x_n), \quad (x_n)_{n \in \mathbb{N}}, \quad (x_n)_{n=1}^{\infty},$$

where $x_n \equiv f(n)$.

Main cases

In this tutorial, the relevant cases are

$$X = \mathbb{R} \quad \text{and} \quad X = \mathbb{R}^m.$$

Recap: Convergence of Sequences

Sequences in $\mathbb{R}$

A sequence (x_n) with codomain $\mathbb{R}$ converges to $x^* \in \mathbb{R}$ if

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N : |x_n - x^*| < \varepsilon.$$

If no such x^* exists, the sequence diverges.

Sequences in $\mathbb{R}^m$

A sequence $(\mathbf{x}_n)$ with codomain $\mathbb{R}^m$ converges to $\mathbf{x}^* \in \mathbb{R}^m$ if

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N : \|\mathbf{x}_n - \mathbf{x}^*\| < \varepsilon.$$

Recap: Open Balls

Definition

For $r > 0$ and center $\mathbf{x} \in \mathbb{R}^m$, the open ball is

$$B_r(\mathbf{x}) = \{\mathbf{y} \in \mathbb{R}^m : \|\mathbf{y} - \mathbf{x}\| < r\}.$$

Convergence using open balls

A sequence $(\mathbf{x}_n)$ converges to $\mathbf{x}^*$ if and only if

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N : \mathbf{x}_n \in B_\varepsilon(\mathbf{x}^*).$$

Recap: Useful Theorems for Sequences

Limit rules in $\mathbb{R}^m$

If $\mathbf{x}_n \rightarrow \mathbf{x}^*$, $\mathbf{y}_n \rightarrow \mathbf{y}^*$, and $c \in \mathbb{R}$, then

$$\mathbf{x}_n + \mathbf{y}_n \rightarrow \mathbf{x}^* + \mathbf{y}^*, \quad c\mathbf{x}_n \rightarrow c\mathbf{x}^*.$$

Componentwise convergence

For $\mathbf{x}_n = (x_{1,n}, \dots, x_{m,n}) \in \mathbb{R}^m$,

$$\mathbf{x}_n \rightarrow \mathbf{x}^* \iff x_{i,n} \rightarrow x_i^* \text{ for every } i = 1, \dots, m.$$

Real product rule

If $x_n \rightarrow x^*$ and $y_n \rightarrow y^*$ in $\mathbb{R}$, then

$$x_n y_n \rightarrow x^* y^*.$$

Recap: Interior, Exterior, Boundary

Definitions for $S \subseteq \mathbb{R}^m$

A point $\mathbf{x} \in \mathbb{R}^m$ is

- ▶ an **interior point** of S if some open ball centered at $\mathbf{x}$ is contained entirely in S ;
- ▶ an **exterior point** of S if some open ball centered at $\mathbf{x}$ is contained entirely outside S ;
- ▶ a **boundary point** of S if every open ball centered at $\mathbf{x}$ contains both a point in S and a point outside S .

Interior of a set

$$\text{int}(S) = \{\mathbf{x} \in S : \mathbf{x} \text{ is an interior point of } S\}.$$

Recap: Open, Closed, Bounded, Compact

Open and closed

A set $S \subseteq \mathbb{R}^m$ is

- ▶ **open** if all of its points are interior points;
- ▶ **closed** if it contains all of its boundary points.

$$S \text{ is open} \iff S = \text{int}(S).$$

Note. A set could be both open and closed; a set could also be both not open and not closed.

Bounded and compact

A set $S \subseteq \mathbb{R}^m$ is bounded if

$$\exists K > 0 \text{ s.t. } \forall \mathbf{x} \in S : \|\mathbf{x}\| \leq K.$$

A set is **compact** if it is closed and bounded.

Question 1

Prove that every open ball is an open set.

Hint. Consider an arbitrary open ball $B_\varepsilon(\mathbf{x}_0)$. For any $\mathbf{x} \in B_\varepsilon(\mathbf{x}_0)$, choose some $\delta > 0$ such that $B_\delta(\mathbf{x}) \subseteq B_\varepsilon(\mathbf{x}_0)$.

Question 1

Prove that every open ball is an open set.

Hint. Consider an arbitrary open ball $B_\varepsilon(\mathbf{x}_0)$. For any $\mathbf{x} \in B_\varepsilon(\mathbf{x}_0)$, choose some $\delta > 0$ such that $B_\delta(\mathbf{x}) \subseteq B_\varepsilon(\mathbf{x}_0)$.

Knowledge used

- Definition of an open ball:

$$B_r(\mathbf{x}) = \{\mathbf{y} \in \mathbb{R}^m : \|\mathbf{y} - \mathbf{x}\| < r\}.$$

- Definition of an interior point: some open ball around the point lies entirely inside the set.
- Definition of an open set: all points are interior points.
- Triangle inequality:

$$\|\mathbf{a} - \mathbf{c}\| \leq \|\mathbf{a} - \mathbf{b}\| + \|\mathbf{b} - \mathbf{c}\|.$$

Tutorial Q1: Solution

Solution

We show that every point of $B_\varepsilon(\mathbf{x}_0)$ is an interior point.

Pick an arbitrary $\mathbf{x} \in B_\varepsilon(\mathbf{x}_0)$ and define

$$r \equiv \|\mathbf{x} - \mathbf{x}_0\|.$$

Since $\mathbf{x} \in B_\varepsilon(\mathbf{x}_0)$, we have $r < \varepsilon$, so

$$\delta \equiv \varepsilon - r > 0.$$

We claim that

$$B_\delta(\mathbf{x}) \subseteq B_\varepsilon(\mathbf{x}_0).$$

Tutorial Q1: Solution Continued

Solution

Let $\mathbf{x}'$ be arbitrary in $B_\delta(\mathbf{x})$. Then

$$\|\mathbf{x}' - \mathbf{x}\| < \delta = \varepsilon - r.$$

By the triangle inequality,

$$\|\mathbf{x}' - \mathbf{x}_0\| \leq \|\mathbf{x}' - \mathbf{x}\| + \|\mathbf{x} - \mathbf{x}_0\| < (\varepsilon - r) + r = \varepsilon.$$

Hence $\mathbf{x}' \in B_\varepsilon(\mathbf{x}_0)$.

Because $\mathbf{x}'$ is arbitrary,

$$B_\delta(\mathbf{x}) \subseteq B_\varepsilon(\mathbf{x}_0).$$

Thus $\mathbf{x}$ is an interior point, and $B_\varepsilon(\mathbf{x}_0)$ is open. □

Question 2

Determine whether the sequence

$$\left(\frac{n}{n+1} \right)_{n \in \mathbb{N}}$$

converges or diverges. If it converges, compute the limit.

Carefully argue using the formal definition of convergence or results from the lecture.

Question 2

Determine whether the sequence

$$\left(\frac{n}{n+1} \right)_{n \in \mathbb{N}}$$

converges or diverges. If it converges, compute the limit.

Carefully argue using the formal definition of convergence or results from the lecture.

Knowledge used

- Formal convergence in $\mathbb{R}$:

$$x_n \rightarrow x^* \iff \forall \varepsilon > 0 \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N : |x_n - x^*| < \varepsilon.$$

- To prove convergence, first guess the limit, then choose N as a function of ε .

Tutorial Q2: Solution

Solution

The sequence converges to 1.

Let $\varepsilon > 0$. Pick $N \in \mathbb{N}$ such that

$$N > \frac{1}{\varepsilon} - 1,$$

which implies

$$\frac{1}{N+1} < \varepsilon.$$

Then for all $n \geq N$,

$$\left| \frac{n}{n+1} - 1 \right| = \left| \frac{n - (n+1)}{n+1} \right| = \frac{1}{n+1} \leq \frac{1}{N+1} < \varepsilon.$$

Therefore,

$$\frac{n}{n+1} \rightarrow 1.$$

Question 3

For each of the following sequences, determine whether the sequence converges or diverges. If it converges, compute the limit.

(a) $(x, x, x, \dots)$, where $x \in \mathbb{R}$.

(b)

$$\left(\left(\frac{n+2}{n+1}, \frac{3n}{n+1} \right) \right)_{n \in \mathbb{N}}.$$

Carefully argue using the formal definition of convergence or results from the lecture.

Tutorial Q3

Question 3

For each of the following sequences, determine whether the sequence converges or diverges. If it converges, compute the limit.

(a) $(x, x, x, \dots)$, where $x \in \mathbb{R}$.

(b)

$$\left(\left(\frac{n+2}{n+1}, \frac{3n}{n+1} \right) \right)_{n \in \mathbb{N}}.$$

Carefully argue using the formal definition of convergence or results from the lecture.

Knowledge used

- ▶ Formal convergence in $\mathbb{R}$.
- ▶ For a constant sequence, the distance from each term to the proposed limit is zero.
- ▶ Componentwise convergence in $\mathbb{R}^m$:

$$\mathbf{x}_n \rightarrow \mathbf{x}^* \iff x_{i,n} \rightarrow x_i^* \text{ for all } i.$$

Tutorial Q3(a): Solution

Solution

Let $x_n = x$ for all $n \in \mathbb{N}$. We claim that $x_n \rightarrow x$.

For every $\varepsilon > 0$, choose $N = 1$. Then for all $n \geq N$,

$$|x_n - x| = |x - x| = 0 < \varepsilon.$$

Therefore, the sequence converges and

$$(x, x, x, \dots) \rightarrow x.$$

Tutorial Q3(b): First Component

Solution

Consider the first component:

$$\frac{n+2}{n+1}.$$

Let $\varepsilon > 0$. Pick $N \in \mathbb{N}$ such that

$$N > \frac{1}{\varepsilon} - 1.$$

Then for all $n \geq N$,

$$\left| \frac{n+2}{n+1} - 1 \right| = \left| \frac{n+2 - (n+1)}{n+1} \right| = \frac{1}{n+1} \leq \frac{1}{N+1} < \varepsilon.$$

Thus the first component converges to 1.

Tutorial Q3(b): Second Component

Solution

Consider the second component:

$$\frac{3n}{n+1}.$$

Let $\varepsilon > 0$. Pick $N \in \mathbb{N}$ such that

$$N > \frac{3}{\varepsilon} - 1.$$

Then for all $n \geq N$,

$$\left| \frac{3n}{n+1} - 3 \right| = \left| \frac{3n - (3n+3)}{n+1} \right| = \frac{3}{n+1} \leq \frac{3}{N+1} < \varepsilon.$$

Thus the second component converges to 3.

Tutorial Q3(b): Conclusion

Solution

By componentwise convergence,

$$\left(\frac{n+2}{n+1}, \frac{3n}{n+1} \right) \rightarrow (1, 3).$$

Therefore,

$$\left(\left(\frac{n+2}{n+1}, \frac{3n}{n+1} \right) \right)_{n \in \mathbb{N}} \text{ converges to } (1, 3).$$

Question 4

Draw the following sets. Explain whether they are open and whether they are compact, and determine their sets of interior points and boundary points.

- (a) $S_1 = \{(x, y) \in \mathbb{R}^2 : |x| + |y| \leq 1\}$.
- (b) $S_2 = \{(x, y) \in \mathbb{R}^2 : (x-1)^2 + y^2 < 1, x^2 + y^2 \geq 1\}$.
- (c) $S_3 = \{(x, y) \in \mathbb{R}^2 : x+1 < y, x+y \leq 2\}$.
- (d) $S_4 = \{(x, y) \in \mathbb{R}^2 : 1 - x^2 \leq y\}$.

Question 4

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- (b) $S_2 = \{(x, y) \in \mathbb{R}^2 : (x-1)^2 + y^2 < 1, x^2 + y^2 \geq 1\}$.
- (c) $S_3 = \{(x, y) \in \mathbb{R}^2 : x+1 < y, x+y \leq 2\}$.
- (d) $S_4 = \{(x, y) \in \mathbb{R}^2 : 1 - x^2 \leq y\}$.

Knowledge used

- ▶ Interior points, boundary points, and exterior points are defined using open balls.
- ▶ S is open if every point of S is an interior point.
- ▶ S is closed if it contains all boundary points.
- ▶ S is compact if and only if it is closed and bounded.
- ▶ Graphing convention: solid boundary means included; dashed boundary means excluded.

Tutorial Q4(a): Solution

Solution

$$S_1 = \{(x, y) \in \mathbb{R}^2 : |x| + |y| \leq 1\}.$$

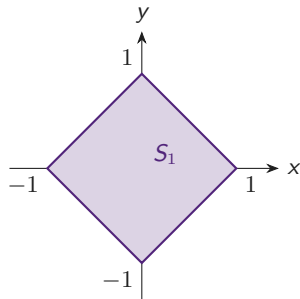
S_1 is not open, but it is compact.

Boundary points:

$$B_1 = \{(x, y) \in \mathbb{R}^2 : |x| + |y| = 1\}.$$

Interior points:

$$I_1 = \{(x, y) \in \mathbb{R}^2 : |x| + |y| < 1\}.$$



Tutorial Q4(b): Solution

Solution

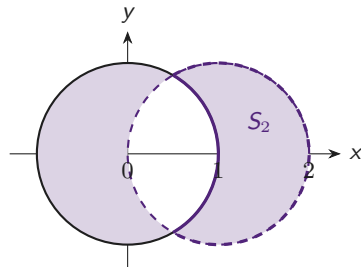
$$\begin{aligned} S_2 &= \{(x, y) \in \mathbb{R}^2 : (x-1)^2 + y^2 < 1, x^2 + y^2 \geq 1\} \\ &= B_1(1, 0) \setminus B_1(0, 0). \end{aligned}$$

Boundary points:

$$\begin{aligned} B_2 &= \{(x, y) : x^2 + y^2 = 1, x \geq \tfrac{1}{2}\} \\ &\quad \cup \{(x, y) : (x-1)^2 + y^2 = 1, x \geq \tfrac{1}{2}\}. \end{aligned}$$

Interior points:

$$I_2 = S_2 \setminus \{(x, y) : x^2 + y^2 = 1, x \geq \tfrac{1}{2}\}.$$



Tutorial Q4(b): Open and Compact?

Solution

- ▶ S_2 is **not closed**: for example, $(2, 0)$ is a boundary point of S_2 , but $(2, 0) \notin S_2$.
- ▶ Since S_2 is not closed, it is **not compact**.
- ▶ S_2 is **not open**: for example, $(1, 0) \in S_2$, but any open ball around $(1, 0)$ contains both points in S_2 and points outside S_2 .

Tutorial Q4(c): Solution

Solution

$$S_3 = \{(x, y) \in \mathbb{R}^2 : x + 1 < y, x + y \leq 2\}.$$

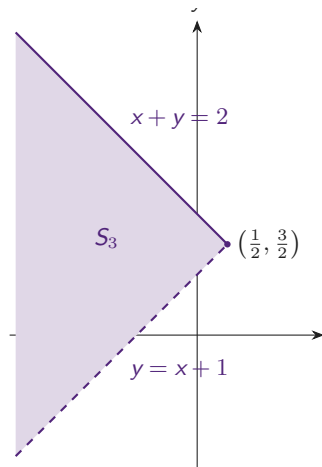
S_3 is neither open nor closed. Since it is not closed, it is not compact. It is also not bounded.

Boundary points:

$$B_3 = \{(x, y) \in \mathbb{R}^2 : x + y = 2, x \leq \tfrac{1}{2}\} \\ \cup \{(x, y) \in \mathbb{R}^2 : x + 1 = y, x \leq \tfrac{1}{2}\}.$$

Interior points:

$$I_3 = \{(x, y) \in \mathbb{R}^2 : x + 1 < y, x + y < 2\}.$$



Tutorial Q4(d): Solution

Solution

$$S_4 = \{(x, y) \in \mathbb{R}^2 : 1 - x^2 \leq y\}.$$

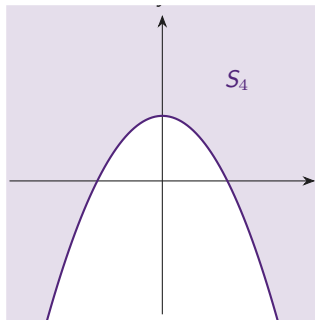
S_4 is not open. It is closed, but it is not bounded, so it is not compact.

Boundary points:

$$B_4 = \{(x, y) \in \mathbb{R}^2 : 1 - x^2 = y\}.$$

Interior points:

$$I_4 = \{(x, y) \in \mathbb{R}^2 : 1 - x^2 < y\}.$$



Question 5

For each of the following functions f , determine the largest domain on which it can be defined as well as the corresponding image of f .

Recall that

$$\sqrt{\cdot} : \mathbb{R}_+ \rightarrow \mathbb{R}, \quad \ln : \mathbb{R}_{++} \rightarrow \mathbb{R},$$

where

$$\mathbb{R}_+ = \{x \in \mathbb{R} : x \geq 0\}, \quad \mathbb{R}_{++} = \{x \in \mathbb{R} : x > 0\}.$$

- (a) $f(x, y) = \sqrt{16 - x^2 - y^2}$.
- (b) $f(x, y) = \sqrt{x^2 + y^2 - 9}$.
- (c) $f(x, y) = \ln [(x^2 - 1)(y - 3)]$.

Tutorial Q5: Knowledge Used

Knowledge used

- ▶ A function is well-defined only on points where its formula makes sense.
- ▶ For square roots, the expression inside must be nonnegative:

$$\sqrt{g(x, y)} \text{ requires } g(x, y) \geq 0.$$

- ▶ For logarithms, the expression inside must be strictly positive:

$$\ln(g(x, y)) \text{ requires } g(x, y) > 0.$$

- ▶ The image is the set of all possible output values.

Tutorial Q5(a): Solution

Solution

$$f(x, y) = \sqrt{16 - x^2 - y^2}.$$

The function is well-defined when

$$16 - x^2 - y^2 \geq 0 \iff x^2 + y^2 \leq 16.$$

Thus the largest domain is

$$D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 16\}.$$

This is the disk with radius 4 and center $(0, 0)$.

Since $16 - x^2 - y^2 \in [0, 16]$, the image is

$$\text{im}(f) = \{z \in \mathbb{R} : 0 \leq z \leq 4\} = [0, 4].$$

Tutorial Q5(b): Solution

Solution

$$f(x, y) = \sqrt{x^2 + y^2 - 9}.$$

The function is well-defined when

$$x^2 + y^2 - 9 \geq 0 \iff x^2 + y^2 \geq 9.$$

Thus the largest domain is

$$D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \geq 9\}.$$

This is the region outside and including the circle with radius 3 and center $(0, 0)$.

The corresponding image is

$$\text{im}(f) = \mathbb{R}_+ = [0, \infty).$$

Solution

$$f(x, y) = \ln [(x^2 - 1)(y - 3)] .$$

The logarithm requires

$$(x^2 - 1)(y - 3) > 0.$$

This product is positive when both factors are positive or both factors are negative. Hence

$$\begin{aligned} D = & \{(x, y) \in \mathbb{R}^2 : x > 1, y > 3\} \\ & \cup \{(x, y) \in \mathbb{R}^2 : x < -1, y > 3\} \\ & \cup \{(x, y) \in \mathbb{R}^2 : -1 < x < 1, y < 3\}. \end{aligned}$$

The corresponding image is

$$\text{im}(f) = \mathbb{R}.$$

Key Takeaways

- ▶ Convergence proofs require choosing N so that the tail is within any ε -distance of the limit.
- ▶ Convergence in $\mathbb{R}^m$ can be checked component by component.
- ▶ Open, closed, bounded, and compact sets are determined by open balls, boundary points, and boundedness.
- ▶ Domains are determined by where formulas are well-defined; images are all possible outputs.